

Elective II

<UEE615 Data Mining and Visualization >				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: What is data mining, data mining objectives, data mining process, input-output data knowledge representation and applications, standard data repositories.				
Syllabus: Overview: What is data mining, data mining objectives, data mining process, input-output data knowledge representation and applications, standard data repositories. Data Pre-processing: Data cleaning, Data transformation, Data reduction, Discretization and generating concept hierarchies, Multidimensional data model Data Analysis and Classification: Attribute generalization and class comparison, Statistical measures, mining through association rules, pattern mining methods, mining diverse frequent patterns, pattern evaluation, sequential pattern mining, graph pattern mining, constraint-based mining and pattern discovery, text mining examples. Data Clustering: Issues in clustering, Partitioning methods: k-means, expectation maximization (EM), Hierarchical distance-based agglomerative and divisible clustering, Data Visualization: Data visualization library and tools, 1D, 2D and 3D charts, Regression plot, Histogram plot, Box and Violin plots, Kernel density estimate plot, Heat maps and clustered matrix, Tree maps; Visualizing large data: Decision Tree Analysis.				
Course Learning Outcomes (CLO) After completion of the course, the students will be able to: 1. Apply basic concepts and techniques of Data Mining. 2. Develop skills of using data mining libraries for solving practical problems. 3. Interpret knowledge by doing hands on session 4.. Test for different data traits				
Text Books 1. Ian H. Witten and Eibe Frank, Data Mining: Practical Machine Learning Tools and Techniques (Second Edition), Morgan Kaufmann, 2005, ISBN: 0-12-088407-0. 2. Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, “Probability & Statistics for Engineers & Scientists”, (9 th Edn.), Prentice Hall Inc. 3. Trevor Hastie Robert Tibshirani Jerome Friedman, “The Elements of Statistical Learning, Data Mining, Inference, and Prediction”, (2 nd Edn.), Springer, 2014.				
Reference Books 1.1G James, D. Witten, T Hastie, and R. Tibshirani, “An Introduction to Statistical Learning: with Applications in R”, Springer, 2013 2. John M. Chambers, “Software for Data Analysis: Programming with R” (Statistics and Computing), Springer 3. Rahlf, Thomas, “Data Visualisation with R”, Springer, 2019.				

Approved in 109th meeting of the Senate held on March 16,2023. Revised in 112th and 114th meeting of the Senate held on March 11,2023 and March 7,2025, respectively.

Evaluation Scheme		
S. No.	Evaluation Elements	Weightage (%)
1.	MST & EST	70
2.	Sessional (Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30

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